



Vibration signal analysis for gear fault diagnosis with various crack progression scenarios

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ABSTRACT

There are different analytical scenarios assumed for crack propagation in the gear tooth root. This paper presents an investigation of the performance of statistical fault detection indicators (the RMS and kurtosis) for three different series of crack propagation scenarios, to compare these scenarios from a fault diagnostics point of view. These scenarios imply different forms of cracks with propagation by a certain step of crack depth. The first scenario assumes a crack being extended through the whole tooth width with a uniform crack depth distribution, while the second scenario assumes the crack being extended through the whole tooth width with a parabolic crack depth distribution, and finally in the third scenario the crack is assumed to be propagating in both the depth and the length directions simultaneously. The time-varying gear mesh stiffness has been investigated using the programme code developed in the present research, and the crack propagation can be modelled with any of the presented crack propagation scenarios. Dynamic simulation has been performed to obtain the residual signals of all the studied cases for each crack propagation scenario. The comparison of the statistical indicators applied to the residual signals shows that in the first scenario the faults are most easily detectable, since in this scenario there is a change in the indicators implying a dramatic decrease in the gear mesh stiffness. The fault detection in the 2nd scenario is more difficult, as the crack propagates with no significant reflection on the mesh stiffness loss. The 3rd proposed scenario should receive more attention in research because it could occur in reality in case of non-uniform load distribution. However, with this scenario it is difficult to perform early fault detection, since there is a very slight change in the statistical indicators at the beginning of the crack propagation. After which, these indicators show a significant change when the crack grows deeper which implies a serious crack propagation condition.

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1. Introduction

Fault diagnosis using vibration analysis is considered as an efficient tool for detecting the faults or damages existing in gears and then giving a warning as early as possible. The role of vibration health monitoring is to detect the deterioration due to faults propagation before the occurrence of sudden breakage. This deterioration results in a malfunction of the

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gearing system and then affects the availability of the whole system. Early detection allows proper scheduled shutdown to prevent catastrophic failure and consequently results in a safer operation and higher cost savings [1,2].

The dynamic behaviour of gear transmissions is affected by operation conditions like excessive applied load, insufficient lubrication, manufacturing errors or installation problems [3]. When an extreme stress concentration is produced in the engaged tooth, a crack may initiate and start propagating during operation. Therefore, for condition monitoring purposes, the status of the tooth damage can be evaluated by the reduction in the time-varying gear mesh stiffness [3–5]. This reduction in the mesh stiffness affects the gear's dynamic behaviour and then the generated vibration [5]. In a study presented by Mark et al. [6] the tooth profile error of a gear was investigated. An increasing profile error could be detected with an increasing crack size. The authors suggested that profile errors due to plastic deformation have a more significant effect on the possibility to detect a gear tooth crack than the effect of stiffness reduction. The effect of adding plastic deformation to dynamic simulations is still to be investigated.

Many research studies have been performed to develop analytical methods for investigating the mesh stiffness of healthy gears [7–11]. These methods are still in use today as their results have shown good agreement with those obtained with FEM, which is a more recent method used for this purpose. FEM is an efficient tool for calculating mesh stiffness [12–14]. It is worth mentioning, however, that FEM needs a great deal of computations, especially in cases which require mesh refinement, so that analytical methods can still be considered as an alternative option for this purpose.

Investigations of gear mesh stiffness have been carried out extensively for uncracked healthy gears, but for faulty gears, crack modelling and mesh stiffness evaluation are fields that are still currently being explored by many researchers [5]. A great deal of research work has been conducted to study the crack propagation path and the effect of the rim thickness on the crack direction [15–17]. It has been found that the path of the crack propagation is dependent on the 'backup ratio', which may be defined as the ratio of the rim thickness to the tooth height. In gears with a high backup ratio, the direction of the crack path in the tooth root will be through the tooth. In gears with a low backup ratio, the crack path will be through the rim. The initial crack angle also affects the crack propagation path in such a way that, when the initial angle is low, the path of the crack may go through the rim, even in the case of a high backup ratio. Investigations of the reduction in the mesh stiffness due to damage or crack propagation have been carried out [18–20]. The fact that the mesh stiffness changes proportionally with the crack status could be used for severity assessment, but these investigations do not define the crack sizes which correspond to the amounts of stiffness reduction found.

Calculation of the total gear mesh stiffness using crack modelling has been introduced through studying the effect of the crack propagation size on the mesh stiffness [21,22]. Wu et al. [22] studied the effect of crack propagation in the tooth root on the dynamic response of a gearbox. The crack levels were simulated from 0% to 80% of the theoretical through-tooth crack size. The resulting vibration signals were analysed by applying several statistical indicators. By comparing the performance of the calculated indicators, the authors concluded that using the residual signal obtained by removing the regular signal of the healthy case from the signal of the faulty case is more sensitive for fault detection. The RMS is the best indicator when using this method of generating the residual signal, and kurtosis is the robust indicator for all the kinds of signals used. Chaari et al. [3] presented an analytical approach for calculating the reduction in the total gear mesh stiffness due to crack depth propagation, as well as a model using FEM to verify the results obtained analytically. An analytical method was proposed by Chaari et al. [5] to evaluate the reduction of the gear mesh stiffness due to spalling and breakage. They obtained the dynamic response corresponding to each fault which was caused by spalling and breaking and which was affected by the reduction of mesh stiffness. The performance of the statistical indicators (the RMS and kurtosis) in combination with application of the discrete wavelet transform (DWT) was studied in [23], with a view of improving the fault detection sensitivity. A modified mathematical model of crack growth in the tooth root was proposed in [24] for calculating time-varying mesh stiffness using an improved potential energy method. This study was carried out using a 16 DOF dynamic model and showed good performance for the applied signal processing technique.

The empirical mode decomposition (EMD) method has been applied in many research studies [25–28]. This technique performs well and is based on decomposing the complicated signal into a set of elementary signals called intrinsic mode functions (IMFs). A technique of differential diagnosis was presented by Endo et al. [29] for diagnosing and recognising two types of gear faults: spalling and cracking. They studied the effects of spalls and cracks on the dynamic response and presented observations for diagnosing the fault type.

In the above-mentioned studies that have dealt with analytical stiffness modelling and calculation, the crack is assumed to extend through the whole tooth width with a uniform crack depth distribution along the tooth width.

Chen and Shao [4] presented an analytical approach for modelling crack propagation with a non-uniform parabolic distribution and then studied the effect of this propagation on the time-varying gear mesh stiffness and the dynamic response. They studied the effect of the size of a crack propagating along the tooth width and the crack depth on the statistical indicators (the RMS and kurtosis). An analytical crack propagation scenario is proposed in [30] which assumes that a crack propagates in the tooth root in both the crack depth direction and the tooth width direction simultaneously, and which is more reasonable and realistic for non-uniform load distribution cases than the other presented scenarios.

Based on the above-mentioned research, one can conclude that more than one scenario have been assumed when applying analytical methods to the investigation of crack propagation in the tooth root. The scenario of a constant or uniformly distributed crack depth through the whole tooth width was assumed first in many research studies [3,21,22,24], after which two more scenarios were presented by Chen and Shao [4]. The first assumed a constant crack depth with a varying crack length and a parabolic distribution along the tooth width, while the second assumed a constant crack length

with a varying crack depth and a parabolic distribution. Ref. [30] assumed that a crack propagates in the tooth root in both the crack depth direction and the tooth width direction simultaneously. Then the authors directed their attention to the possibility of conducting further required research to compare and evaluate all the presented crack propagation scenarios from the condition monitoring point of view.

In reality there are many different crack propagation scenarios. The difference between these scenarios regarding the fault detection possibility has not been investigated in the previous research. The current paper therefore presents a comparative study of the performance of the statistical indicators (the RMS and kurtosis) for three crack propagation scenarios when performing vibration-based fault detection. This paper uses a previously presented analytical approach [4] for evaluating the time-varying gear mesh stiffness required for obtaining the dynamic response of three series of crack propagation scenarios. However, the crack model used is a strong simplification of a real crack and will overestimate the influence on the stiffness. But, it is essential to use the simplified model to evaluate the analysis method against earlier studies.

2. Gear mesh stiffness modelling with a crack in the tooth root

Gear mesh stiffness can be explained as a factor describing the stiffness produced between two engaged gears due to the stiffness of the meshing teeth of both the gears. Gear mesh stiffness is a time-varying factor during the operation of gears and is dependent on the number of engaged teeth, the applied load's position and direction, the tooth geometry, the gear's material specifications, and the profile errors and/or faults in the gear tooth [3].

An important observation was reported by Lewicki [15], namely that the paths of the crack depth propagation are smooth, continuous, and tend to be straight with a slight curvature. Therefore, the crack propagation path is assumed to be a straight line in the crack depth direction in [3,4,22]. In the crack modelling of the current study, the crack depth is simulated as propagating in a straight line.

A calculation model for gear mesh stiffness using a parabolic distribution of the tooth crack propagation is presented in [4] and can be described as consisting of two parts: determination of the mesh stiffness with a constant crack depth for a thin slice along the tooth width and determination of the mesh stiffness for all the slices along the tooth width which have a non-uniform crack depth. Moreover, the effects of fillet foundation deflection and Hertzian contact are taken into account, as explained in Sections 2.3 and 2.4.

2.1. The mesh stiffness with a constant crack depth for a slice along the tooth width

By considering the tooth as a non-uniform cantilever beam with an effective length of 'd', see Fig. 1d, the deflections under the action of the force can be determined, and then the stiffness can be calculated. Note that, in this part of the stiffness calculations, the crack is assumed to have a constant crack depth, $q(z)$, for each slice, dW , along the tooth width, as explained in Fig. 1a and b.

Based on the calculation of the potential energy stored in a meshing gear tooth, it is feasible to obtain the bending, shear, and axial compressive stiffnesses as follows [4]:

$$\frac{1}{K_b} = \int_0^d \frac{(x \cos(\alpha_1) - h \sin(\alpha_1))^2}{EI_x} dx \quad (1)$$

$$\frac{1}{K_s} = \int_0^d \frac{1.2 \cos^2(\alpha_1)}{GA_x} dx \quad (2)$$

$$\frac{1}{K_a} = \int_0^d \frac{\sin^2(\alpha_1)}{EA_x} dx \quad (3)$$

where the following notation is used:

K_b : Bending stiffness.

K_s : Shear stiffness.

K_a : Axial compressive stiffness.

h , h_c , h_x , x , dx , d , and α_1 are illustrated in Fig. 1d. Note that α_1 varies with the gear tooth position.

E : Young's modulus.

G : Shear modulus,

$$G = \frac{E}{2(1+\nu)}$$

ν : Poisson ratio.

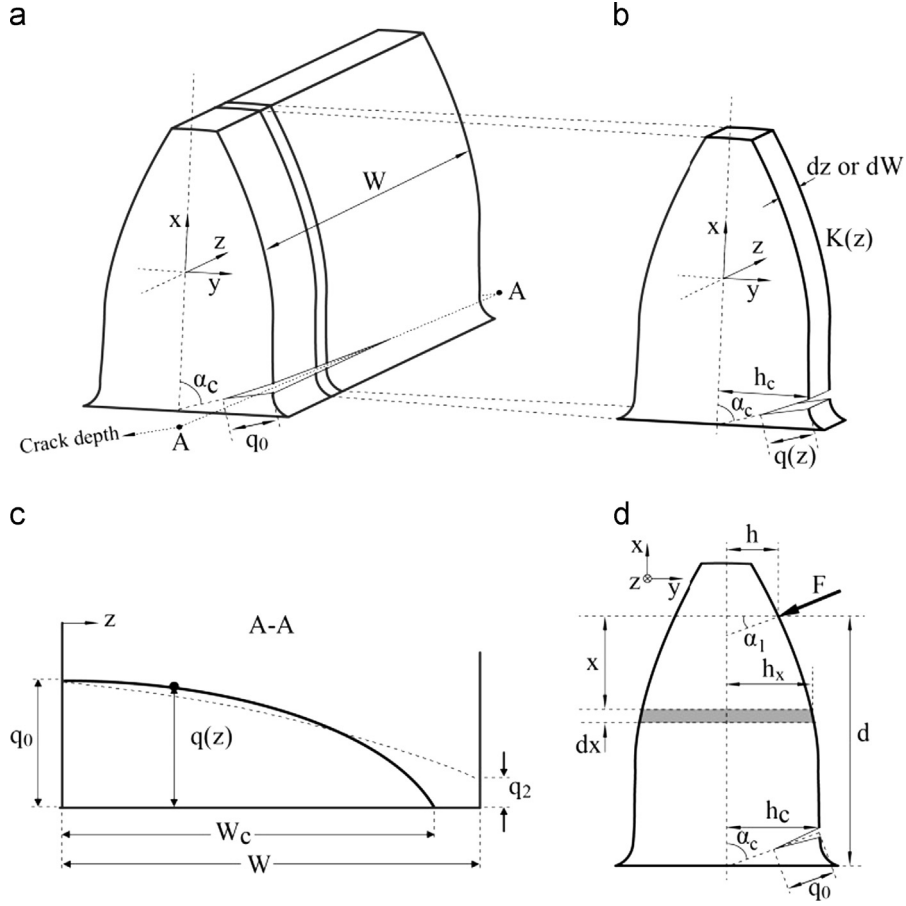


Fig. 1. Modelling of gear tooth crack. (a) Modelling of cracked tooth, (b) modelling of one slice of width dW , (c) crack depth distribution along the tooth width, and (d) tooth notation [4].

I_x : Area moment of inertia,

$$I_x = \begin{cases} (1/12)(h_x + h_q)^3 dW, & h_x \leq h_q \\ (1/12)(h_x + h_q)^3 dW, & h_x > h_q \end{cases}$$

A_x : Area of the section of distance 'x' measured from the load application point,

$$A_x = \begin{cases} (h_x + h_q) dW, & h_x \leq h_q \\ (h_x + h_q) dW, & h_x > h_q \end{cases}$$

$h_q = h_c - q(z) \sin(\alpha_c)$, h_q represents the reduced dimension of the tooth thickness when a crack is existing.

Chen and Shao [4] and Tian [21] used this reduced dimension for calculating I_x and A_x . They used h_q instead of h_x in case of $h_x > h_q$.

$q(z)$ and α_c are the crack depth and crack angle, respectively, as shown in Fig. 1b.

At a certain position, 'z', through the tooth width, we can find the stiffness of one slice resulting from the effect of all the stiffnesses calculated previously as follows:

$$K(z) = 1 / \left(\frac{1}{K_b} + \frac{1}{K_s} + \frac{1}{K_a} \right) \quad (4)$$

2.2. The mesh stiffness for all the slices along the tooth width

The mesh stiffness model presented in [4] divides the tooth width into thin slices to represent the crack propagation through the tooth width, as shown in Fig. 1a. Consequently, for a small dW the crack depth is assumed to be a constant through the width for each slice, see Fig. 1b.

By integrating the stiffness of all the slices along the width, the stiffness of the entire tooth can be evaluated as follows:

$$K_{tp} = \int_0^W K(z) \quad (5)$$

In [4] it is assumed that the distribution of the crack depth follows a parabolic function along the tooth width, as shown in Fig. 1c for the crack section A-A, which can be recognised in Fig. 1a.

When the crack length is less than the whole tooth width:

$$q(z) = q_o \sqrt{\frac{W_c - z}{W_c}}, \quad z \in [0 - W_c] \quad (6)$$

$$q(z) = 0, \quad z \in [W_c - W] \quad (7)$$

where W_c is the crack length, W is the whole tooth width, and q_o is the maximum crack depth, see Fig. 1c.

When the crack length extends through the whole tooth width:

$$q(z) = \sqrt{\frac{q_o^2 - q_2^2}{W} z + q_2^2} \quad (8)$$

where q_2 is shown in Fig. 1c.

2.3. The effect of fillet foundation deflection on the mesh stiffness

Sainsot et al. [31] studied the effect of fillet foundation deflection on the gear mesh stiffness, derived this deflection, and applied it for a gear body. The fillet foundation deflection can be calculated as follows [31]:

$$\delta_f = \frac{F \cdot \cos^2(\alpha_m)}{W \cdot E} \left\{ L^* \left(\frac{u_f}{S_f} \right)^2 + M^* \left(\frac{u_f}{S_f} \right) + P^* (1 + Q^* \tan^2(\alpha_m)) \right\} \quad (9)$$

where the following notation is used:

α_m is the pressure angle.

u_f and S_f are illustrated in Fig. 2.

L^* , M^* , P^* , and Q^* can be approximated using polynomial functions as follows [31]:

$$X_i^*(h_{fi}, \theta_f) = A_i/\theta_f^2 + B_i h_{fi}^2 + C_i h_{fi}/\theta_f + D_i/\theta_f + E_i h_{fi} + F_i \quad (10)$$

X_i^* represents the coefficients L^* , M^* , P^* , and Q^* .

$h_{fi} = r_f/r_{int}$.

r_f , r_{int} , and θ_f are illustrated in Fig. 2.

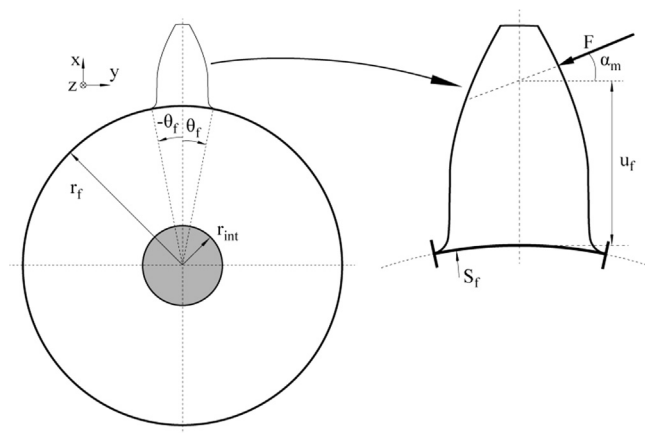


Fig. 2. Geometrical parameters for the fillet foundation deflection [3].

The coefficients A_i , B_i , C_i , D_i , E_i and F_i are given in Table 1.

Then the stiffness due to fillet foundation deflection can be obtained as follows:

$$\frac{1}{K_f} = \frac{\delta_f}{F} \quad (11)$$

For a pinion it could be denoted as K_{fp} .

2.4. The effect of Hertzian contact on the mesh stiffness

Yang and Sun [32] ascertained that the stiffness of the Hertzian contact of two gears in mesh is constant during the whole contact period, and therefore, has the same value at all the contact positions along the path of contact. The Hertzian contact stiffness K_h can be calculated as follows:

$$\frac{1}{K_h} = \frac{4(1-\nu^2)}{\pi \cdot E \cdot W} \quad (12)$$

3. Total mesh stiffness

After calculating the stiffness of a cracked pinion tooth, K_{tp} , due to bending, shear, and axial compression and then calculating the stiffness due to the fillet foundation deflection, K_{fp} , we can perform the same calculations for an uncracked mating gear tooth to find K_{tg} and K_{fg} . Thus, the total mesh stiffness for one meshing tooth pair is as follows:

$$K_t = \frac{1}{\frac{1}{K_{tp}} + \frac{1}{K_{fp}} + \frac{1}{K_{tg}} + \frac{1}{K_{fg}} + \frac{1}{K_h}} = K_{m1} \quad (13)$$

where K_{m1} is the total mesh stiffness for the first tooth pair.

In cases where there are two tooth pairs in contact, the same calculations are repeated for the second tooth pair to find K_{m2} . Then we can obtain the equivalent mesh stiffness as follows:

$$K_m = K_{m1} + K_{m2} \quad (14)$$

In the mesh stiffness calculation, one needs to know the load position relative to the teeth in contact along the path of contact. Note that the faulty tooth position should be changed through the entire contact length for investigating the time-varying mesh stiffness. To obtain the locations of the contact points along the path of contact, the contact zones can be determined from the contact ratio and the base circular pitch [33].

Basically,

$$\text{C.R.} = \frac{\text{PTH}}{P_b} \quad (15)$$

where C.R. is the contact ratio, PTH is the path of contact, and P_b is the base circular pitch.

The load transmitted by low contact ratio gearing is transmitted by a single pair of teeth for a part of the engagement period and by two pairs of teeth during the rest of the period. At the beginning of contact there are two pairs in contact continuing to mesh for a specified period, and then one pair go out of contact, while the other pair continue to mesh alone, until a new pair come into action [34]. As an example, a contact ratio of (1.6) means that the path of contact is equal to 1.6 times the base circular pitch. The path of contact is divided into three zones, as shown in Fig. 3a, and at the beginning of the path of contact, there are two pairs continuing to mesh for 60% of the base circular pitch along the path of contact. Then one pair go out of contact and the other pair continue to mesh alone for the remaining 40% of the base circular pitch, after which another new pair come into contact for a further 60% of the base circular pitch. In general, the contact zones along the line of action for a low contact ratio can be found as described in Fig. 3b [35].

Table 1
Values of the coefficients of Eq. (10).

	$A_i (\times 10^{-5})$	$B_i (\times 10^{-3})$	$C_i (\times 10^{-4})$	$D_i (\times 10^{-3})$	E_i	F_i
$L^* (h_{fn}, \theta_f)$	−5.574	−1.9986	−2.3015	4.7702	0.0271	6.8045
$M^* (h_{fn}, \theta_f)$	60.111	28.100	−83.431	−9.9256	0.1624	0.9086
$P^* (h_{fn}, \theta_f)$	−50.952	185.50	0.0538	53.300	0.2895	0.9236
$Q^* (h_{fn}, \theta_f)$	−6.2042	9.0889	−4.0964	7.8297	−0.1472	0.6904

4. One-stage gear modelling

A single-stage spur gear model has been adopted in the present research work for simulations performed to investigate the time-varying gear mesh stiffness of two mating gears. The main gear modelling parameters that are used in this paper have been taken from [3] and can be seen in Table 2. Here the term *pinion* refers to the smaller gear, which is a driver gear in this study, and the term *gear* refers to the larger gear, which is a driven gear in this study.

5. Computer simulation of time-varying gear mesh stiffness

A programme has been made using Matlab code to determine the (x,y) coordinates of the points forming the involute tooth profile which are required in the mesh stiffness calculation. These tooth profile coordinates can be obtained for different gear dimensions, and also there is the possibility of finding the profile coordinates for a modified long addendum gear tooth.

AutoCAD software has been used to simulate the tooth action along the path of contact. Thus, all the required parameters for mesh stiffness calculation can be obtained.

A programme has been developed using Matlab code to investigate the time-varying mesh stiffness for two gears in contact. The mesh stiffness has been calculated point by point during the time of teeth engagement. The stiffness of a

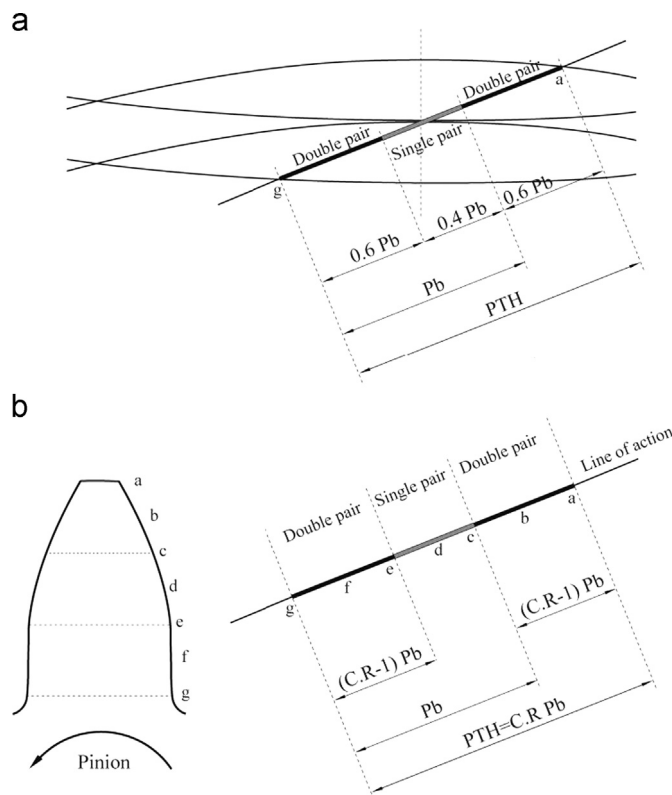


Fig. 3. Contact zones along the path of contact. (a) Contact zones for a contact ratio of 1.6 and (b) contact zones for the low contact ratio gears.

Table 2
Parameters of gear–pinion set [3].

Parameter	Pinion	Gear
Number of teeth	25	30
Module (mm)		2
Teeth width (mm)		20
Contact ratio		1.63
Pressure angle (deg.)		20
Young's modulus, E (N/mm ²)		2×10^5
Poisson's ratio		0.3

cracked tooth with a parabolic crack distribution can be investigated according to the stiffness calculation that is explained in Section 2. With the programme code the crack propagation can be modelled for any one of the presented crack propagation scenarios.

5.1. Programme code validation

To verify the programme code developed in this research study for time-varying gear mesh stiffness, the results obtained with this code have been compared with the results for the studied cases presented in [3], which were obtained using FEM. The parameters used for this comparison (which are also used for gear modelling in this paper) are shown in Table 2 [3]. It is worth mentioning that to fit the comparison of the model used for this comparison considers the larger gear as a driver and the smaller one as a driven gear which is the opposite of the model studied in this paper.

We can explain the cases studied for comparison as follows [3]:

1. Healthy case: no crack.
2. Crack no. 1: $p_c=0.3$ mm, $\alpha_c=33^\circ$ (here we let $q_o=q_2=0.3$ mm and $\alpha_c=33^\circ$ to represent this crack case).
3. Crack no. 2: $p_c=0.66$ mm, $\alpha_c=70^\circ$ (here we let $q_o=q_2=0.66$ mm and $\alpha_c=70^\circ$ to represent this crack case).

If we compare the values of the two result columns in Table 3, we find that they are quite close to each other, and the percentage difference is explained in the last column. Fig. 4 shows the varying mesh stiffness obtained in [3] and that obtained using the programme code developed in the present research work. The comparisons explained in Table 3 and Fig. 4 show good agreement and validate the programme code developed in the present study.

6. Analytical crack propagation scenarios

To study the reduction in the mesh stiffness due to crack propagation in the gear tooth root, several analytical crack propagation scenarios have been proposed. First, the scenario of a constant or uniformly distributed crack depth through the whole tooth width was assumed in many research studies [3,21,22,24], and this assumption is considered to be correct for uniform load distribution cases. After that, two more scenarios with a non-uniform parabolic crack distribution were presented in [4], the first of which assumed a constant crack depth with a varying crack length and a parabolic distribution along the tooth width, and the second assumed a constant crack length with a varying crack depth and a parabolic distribution. A different analytical crack propagation scenario is proposed in [30] which assumes that a crack propagates in the tooth root in both the crack depth direction and the tooth width direction simultaneously, and which is more reasonable and realistic for non-uniform load distribution cases than the other presented scenarios.

The crack level (CL) can be described as a percentage of the total tooth thickness measured at the crack initiation point in the tooth root, which is 3.72 mm in our model. In the present study, three different crack propagation scenarios have been studied and they can be explained as follows:

1. Crack extended through the whole tooth width with a uniform crack depth distribution.
This scenario is considered by [3,21,22,24] and in other research studies mentioned in Section 1. The propagation case data of this scenario are shown in Table 4.
Fig. 5a explains the propagation cases of this scenario, and the time-varying mesh stiffnesses for all the studied cases are shown in Fig. 5b.
2. Crack extended through the whole tooth width with a parabolic crack depth distribution.
This scenario is presented in [4]. The propagation case data of this scenario are shown in Table 5.
Fig. 6a explains the propagation cases of this scenario, and the time-varying mesh stiffnesses for all the studied cases are shown in Fig. 6b.
3. Crack propagating in both the depth and the length directions together.

This scenario is presented in [30]. Practical experience shows that the load distribution along the tooth width is usually non-uniform [36,37], see Fig. 7, with a number of factors causing the deviation from the ideal uniform load distribution, e.g. manufacturing errors, gear shaft misalignment, and deformations in the bearing and/or housing. Experimental tests using a one-sided load distribution have shown that the fatigue crack propagates in two directions: the crack length and the crack

Table 3
Single tooth mesh stiffness comparison.

Crack case	FEM result from [3]	Result from the programme used in this research study	Difference (%)
Healthy case	1.58×10^8	1.57×10^8	0.63
Crack no. 1	1.53×10^8	1.52×10^8	0.65
Crack no. 2	1.42×10^8	1.38×10^8	2.8

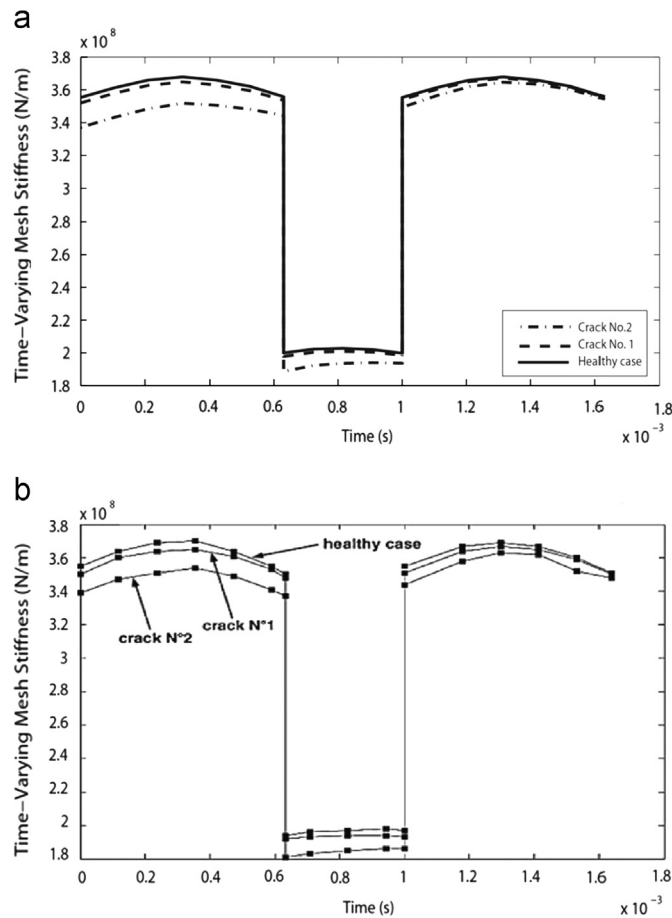


Fig. 4. Time-varying gear-mesh stiffness of the model used for programme code validation. (a) Using the programme developed in this research study and (b) using FEM from [3].

Table 4

Propagation case data of the 1st scenario.

Case	$q_0 = q_2$ (mm)	CL %	Case	$q_0 = q_2$ (mm)	CL %
1	0	0	11	0.9	24.19
2	0.05	1.34	12	1.0	26.88
3	0.1	2.68	13	1.1	29.56
4	0.2	5.37	14	1.2	32.25
5	0.3	8.06	15	1.3	34.94
6	0.4	10.75	16	1.4	37.63
7	0.5	13.44	17	1.5	40.32
8	0.6	16.12	18	1.6	43.01
9	0.7	18.81	19	1.7	45.69
10	0.8	21.50	20	1.8	48.38

$$W_c = W\alpha_c = 70^\circ$$

depth directions [36]. It is more reasonable and realistic for non-uniform load distribution cases to assume that the crack propagates in both the crack depth direction and the crack length direction together. As it is difficult to define analytically the exact crack length corresponding to a certain crack depth, and because the study of crack growth with time is out of the scope of the present research study, we have considered the propagation cases proposed in [30]. The propagation cases utilised for the study of this scenario are shown in Table 6. The values of q_2 have been obtained from the intersections of the parabolic lines with the vertical axis at points a–j, as illustrated in Fig. 8a, the section A–A can be recognised in Fig. 1a, and the time-varying mesh stiffnesses for all the studied cases are shown in Fig. 8b. A high backup ratio with a crack angle of 70° is considered here.

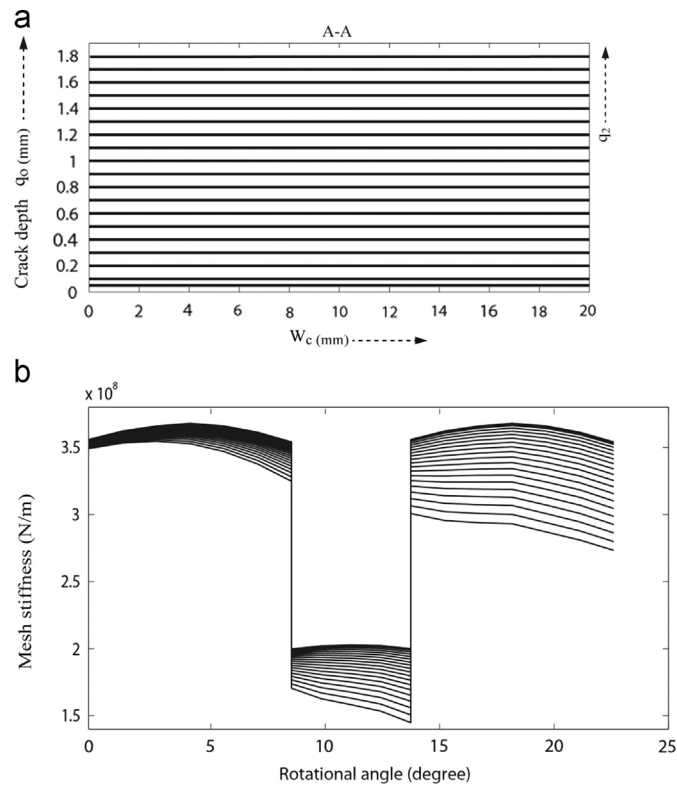


Fig. 5. Crack propagation and time-varying gear mesh stiffness with the 1st scenario. (a) Crack section A-A shows the crack propagation and (b) time-varying gear mesh stiffness.

Table 5

Propagation case data of the 2nd scenario.

Case	q_o (mm)	CL %	Case	q_o (mm)	CL %
1	0	0	11	0.9	24.19
2	0.05	1.34	12	1.0	26.88
3	0.1	2.68	13	1.1	29.56
4	0.2	5.37	14	1.2	32.25
5	0.3	8.06	15	1.3	34.94
6	0.4	10.75	16	1.4	37.63
7	0.5	13.44	17	1.5	40.32
8	0.6	16.12	18	1.6	43.01
9	0.7	18.81	19	1.7	45.69
10	0.8	21.50	20	1.8	48.38

$$W_c = Wq_2 = 0\alpha_c = 70^\circ$$

7. Dynamic modelling

The dynamic response of a gear system can be extracted using dynamic lumped parameters modelling to study the effect of tooth crack propagation on the obtained vibration response from a fault detection point of view. A dynamic simulation of a six DOF model has been performed based on the time-varying mesh stiffness model which was explained earlier. Fig. 9 shows the dynamic model which was used in the present research study and which was adopted in [4,38,39]. This model represents six DOF and is explained in the equations below by introducing the friction force caused by sliding between the mating teeth. The parameters of the dynamic model are explained in Table 7. The gear system works under a torque of 60 N m applied on the driven gear. It is assumed that the radial stiffness and damping in the bearings of the pinion and gear have the same values as those given in Table 7.

The equations of motion in the 'y' direction for the pinion and gear are as follows:

$$m_p \ddot{y}_p = -N - k_{yp} y_p - c_{yp} \dot{y}_p \quad (16)$$

$$m_g \ddot{y}_g = N - k_{yg} y_g - c_{yg} \dot{y}_g \quad (17)$$

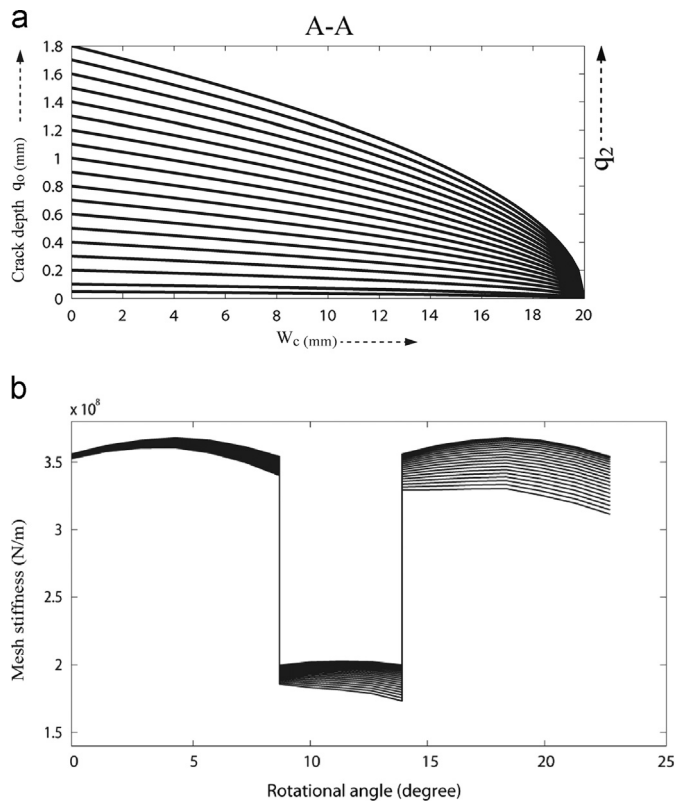


Fig. 6. Crack propagation and time-varying gear mesh stiffness with the 2nd scenario. (a) Crack section A-A shows the crack propagation and (b) time-varying gear mesh stiffness.

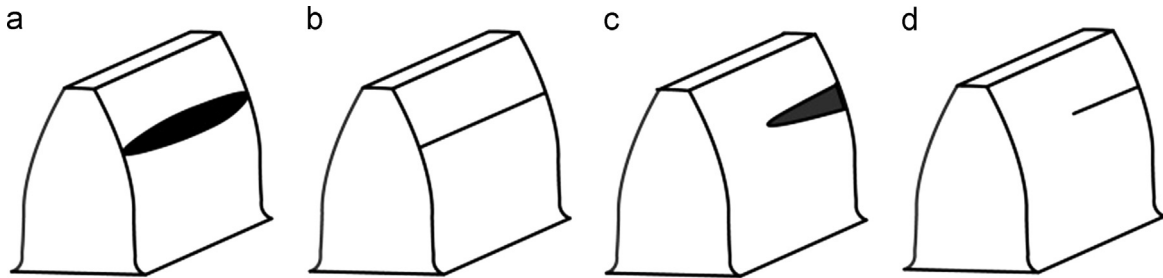


Fig. 7. Illustration of different load distribution cases. (a) Uniform load distribution, (b) uniform load distribution represented analytically, (c) non-uniform or one-sided load distribution and (d) non-uniform load distribution represented analytically.

Table 6
Data for the propagation cases of the proposed scenario.

Case	q_0 (mm)	CL %	W_c (mm)	q_2 (mm)	Case	q_0 (mm)	CL %	W_c (mm)	q_2 (mm)
1	0	0	0	0	11	0.9	24.19	20.0	0.23
2	0.05	1.34	1.25	0	12	1.0	26.88	20.0	0.45
3	0.1	2.68	2.5	0	13	1.1	29.56	20.0	0.575
4	0.2	5.37	5.0	0	14	1.2	32.25	20.0	0.7
5	0.3	8.06	7.5	0	15	1.3	34.94	20.0	0.8125
6	0.4	10.75	10.0	0	16	1.4	37.63	20.0	0.925
7	0.5	13.44	12.5	0	17	1.5	40.32	20.0	1.0325
8	0.6	16.12	15.0	0	18	1.6	43.01	20.0	1.14
9	0.7	18.81	17.5	0	19	1.7	45.69	20.0	1.2475
10	0.8	21.50	20.0	0	20	1.8	48.38	20.0	1.355

$\alpha_c = 70^\circ$

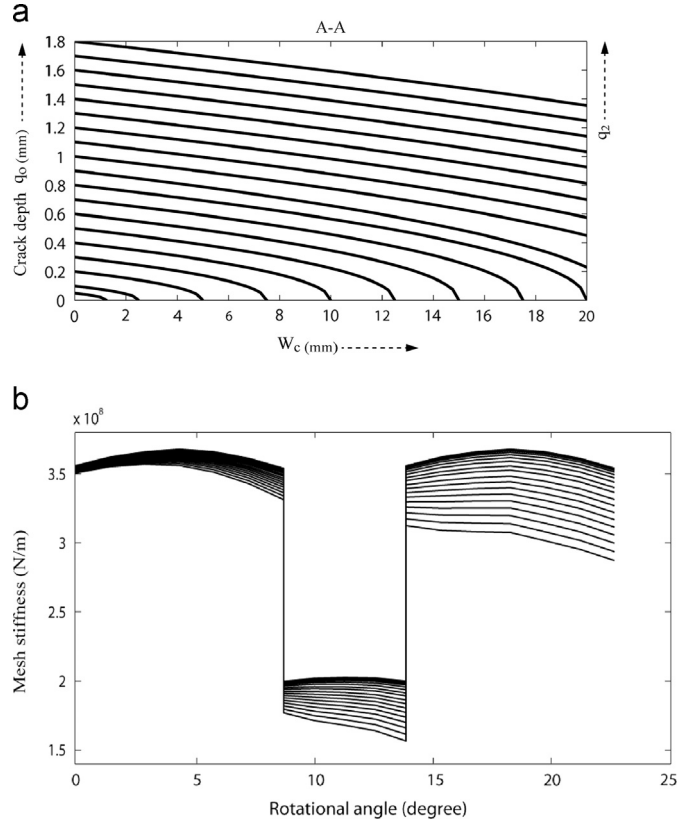


Fig. 8. Crack propagation and time-varying gear mesh stiffness with the proposed 3rd scenario. (a) Crack section A-A shows the crack propagation and (b) time-varying gear mesh stiffness.

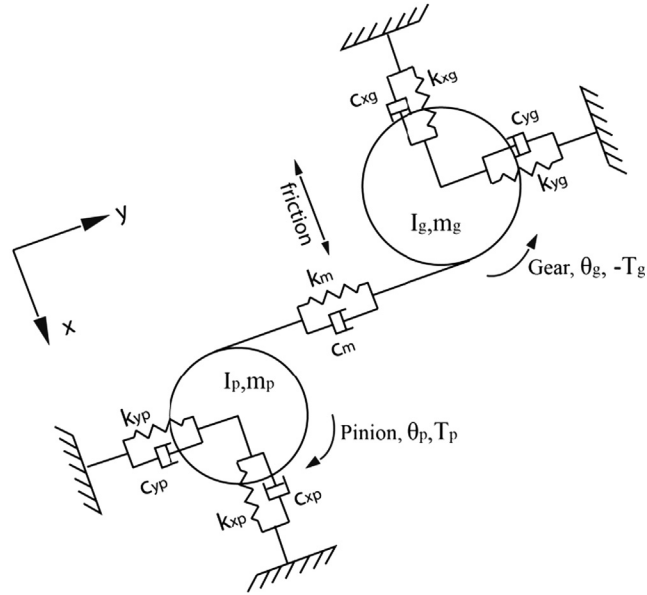


Fig. 9. Dynamic model of a reduction gear system with six DOF.

The equations of motion in the rotary ' θ ' direction for the pinion and gear are as follows:

$$I_p \ddot{\theta}_p = r_p N + T_p + M_p \quad (18)$$

Table 7
Parameters of gear–pinion dynamic model [4].

Parameter	Pinion	Gear
Mass (kg)	0.3083	0.4439
Mass moment of inertia (kg m ²)	0.96e−4	2e−4
Radial stiffness of the bearing in x, y directions (N/m)	6.56e8	6.56e8
Radial damping of the bearing in x, y directions (N s/m)	1.8e3	1.8e3
Rotational speed (rpm)	2400	2000
Gear mesh frequency (Hz)		1000
Coefficient of friction		0.06
Total damping between meshing teeth (N s/m)		67

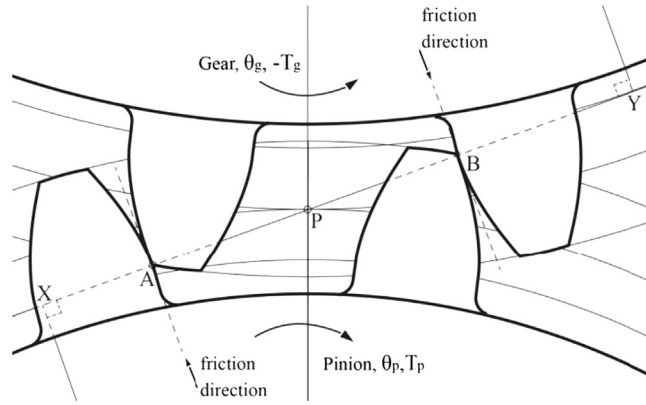


Fig. 10. Friction forces in meshed teeth.

$$I_g \ddot{\theta}_g = -r_g N - T_g + M_g \quad (19)$$

The equations of motion in the 'x' direction for the pinion and gear are as follows:

$$m_p \ddot{x}_p = -k_{xp} x_p - c_{xp} \dot{x}_p + F_p \quad (20)$$

$$m_g \ddot{x}_g = -k_{xg} x_g - c_{xg} \dot{x}_g + F_g \quad (21)$$

where

$$N = k_m [(y_p - y_g) - (r_p \theta_p - r_g \theta_g)] + c_m [(\dot{y}_p - \dot{y}_g) - (r_p \dot{\theta}_p - r_g \dot{\theta}_g)] N = N_1 + N_2 \cdots N_n$$

where n is the number of teeth in contact

$$F_p = F_{p1} + F_{p2} \cdots F_{pn} F_{p1} = \mu N_1 \quad \text{and} \quad F_{p2} = -\mu N_2 \quad F_g = F_{g1} + F_{g2} \cdots F_{gn} F_{g1} = -\mu N_1 \quad \text{and} \quad F_{g2} = \mu N_2$$

We can obtain the moments due to the friction force M_p and M_g by multiplying F_p and F_g , respectively, by their arms at the point of contact, see Fig. 10. The points of contact A and B can be measured from points X and Y to obtain these arms. The following notation is used in the dynamic model equations:

m_p/m_g : Mass of the pinion/gear.

I_p/I_g : Mass moment of inertia of the pinion/gear.

k_{px}/k_{py} : Radial stiffness in the x/y directions of the pinion.

k_{gx}/k_{gy} : Radial stiffness in the x/y directions of the gear.

c_{px}/c_{py} : Radial damping in the x/y directions of the pinion.

c_{gx}/c_{gy} : Radial damping in the x/y directions of the gear.

k_m : Equivalent mesh stiffness.

c_m : Mesh damping coefficient.

T_p/T_g : Torque applied on the pinion/gear.

r_{bp}/r_{bg} : Base circle radius of the pinion/gear.

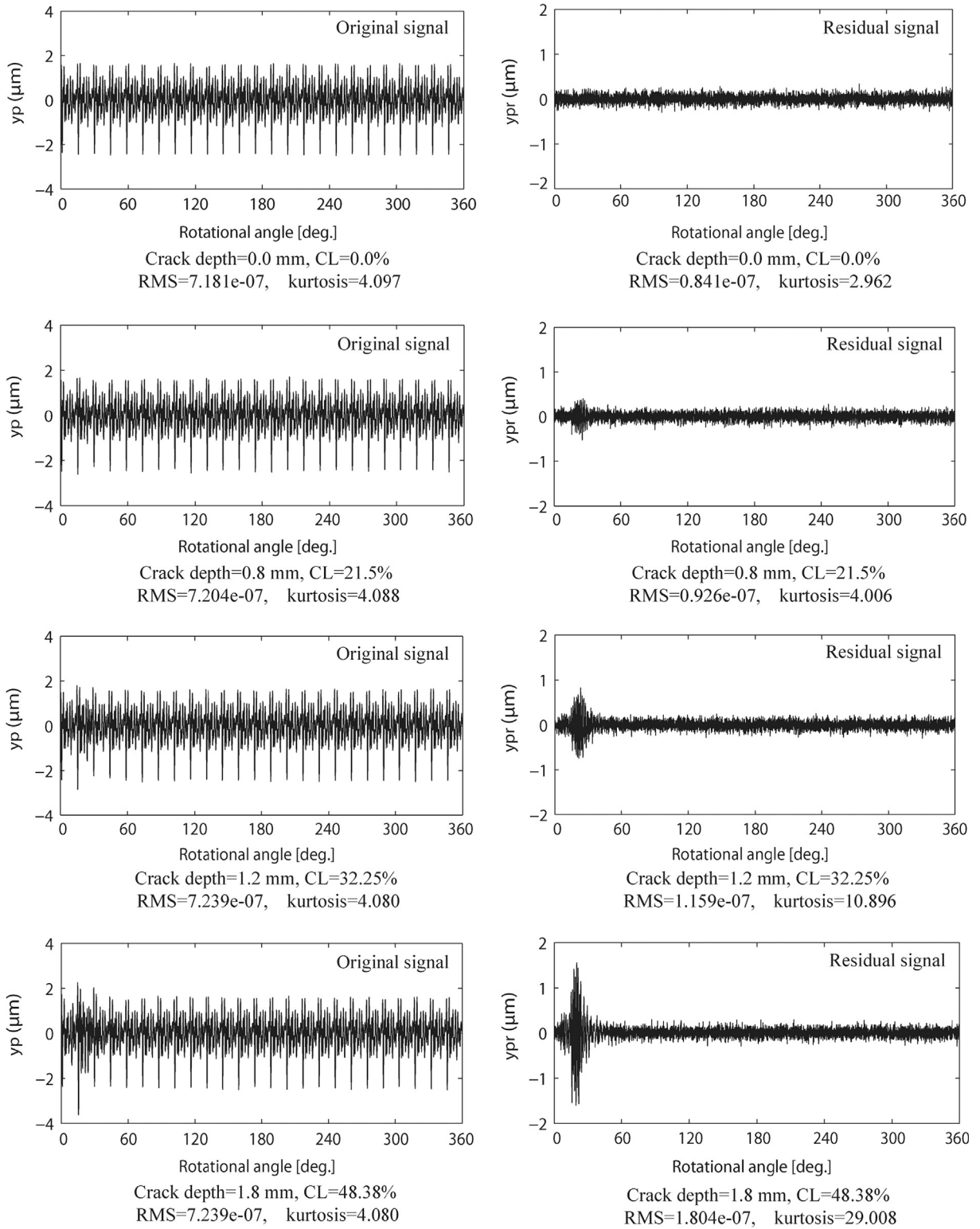


Fig. 11. Selected original signals and their residual signals for the 3rd scenario with the RMS and kurtosis values specified.

A Matlab computer simulation using the ODE45 function was used for modelling the equations of motion with a sampling frequency of 200 kHz to prevent aliasing of the highest detectable harmonics at 20 kHz. The dynamic simulation was performed for the healthy case, after which the simulation was repeated for the faulty cases corresponding to the crack

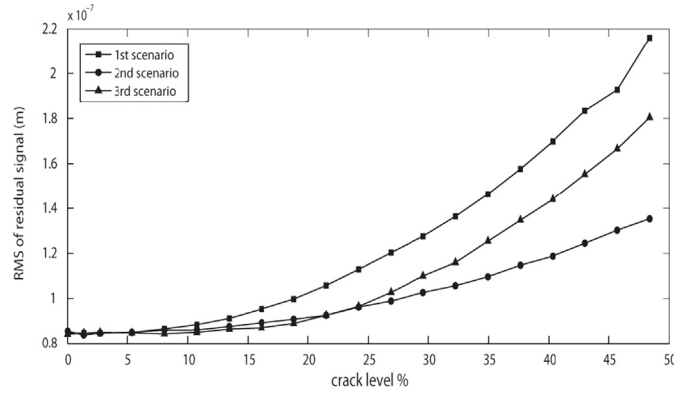


Fig. 12. RMS change for three crack propagation scenarios.

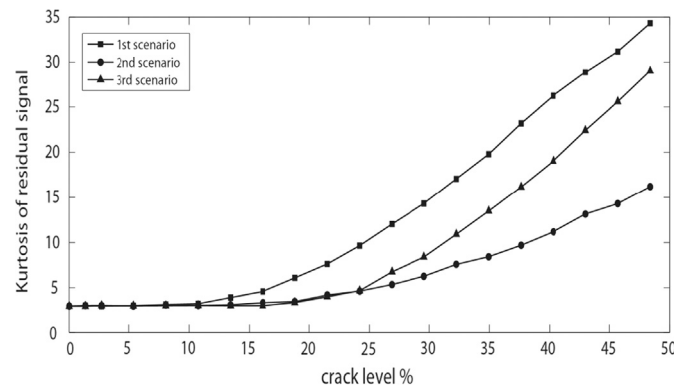


Fig. 13. Kurtosis change for three crack propagation scenarios.

sizes. In the present research study, three crack propagation scenarios were studied, with 20 cases studied for each scenario, as explained in Section 6. This means that, to complete the investigation, 60 cases were simulated to extract the dynamic behaviour.

8. Statistical fault diagnosis indicators

To assess the fault condition, statistical indicators are commonly applied to the obtained signal that describes the system's dynamic behaviour. These indicators permit efficient fault detection, which is necessary to prevent a sudden breakage in the machinery system.

Wu et al. [22] studied the performance of some of the statistical indicators which are applied to different kinds of signals. The conclusion drawn from their study was that the RMS indicator shows the best performance when the 2nd proposed method of generating the residual signal is applied. Kurtosis is the most robust indicator for all the signals used. The 2nd method which they recommended for generating the residual signal involves removing the whole original signal of the healthy case from the original signal of the faulty cases. The original signal which is obtained when there is no crack in the gear teeth is considered as a regular signal representing the healthy case. By removing this regular signal from the faulty one, the influence of regular vibration can be removed and then the signal components generated due to crack propagation can be highlighted. The above-explained method for generating the residual signal has been applied in the present research study.

The RMS is considered as one of the basic statistical indicators that measure the energy level of a signal. The RMS can be defined as follows [1]:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{n=1}^N (x(n))^2} \quad (22)$$

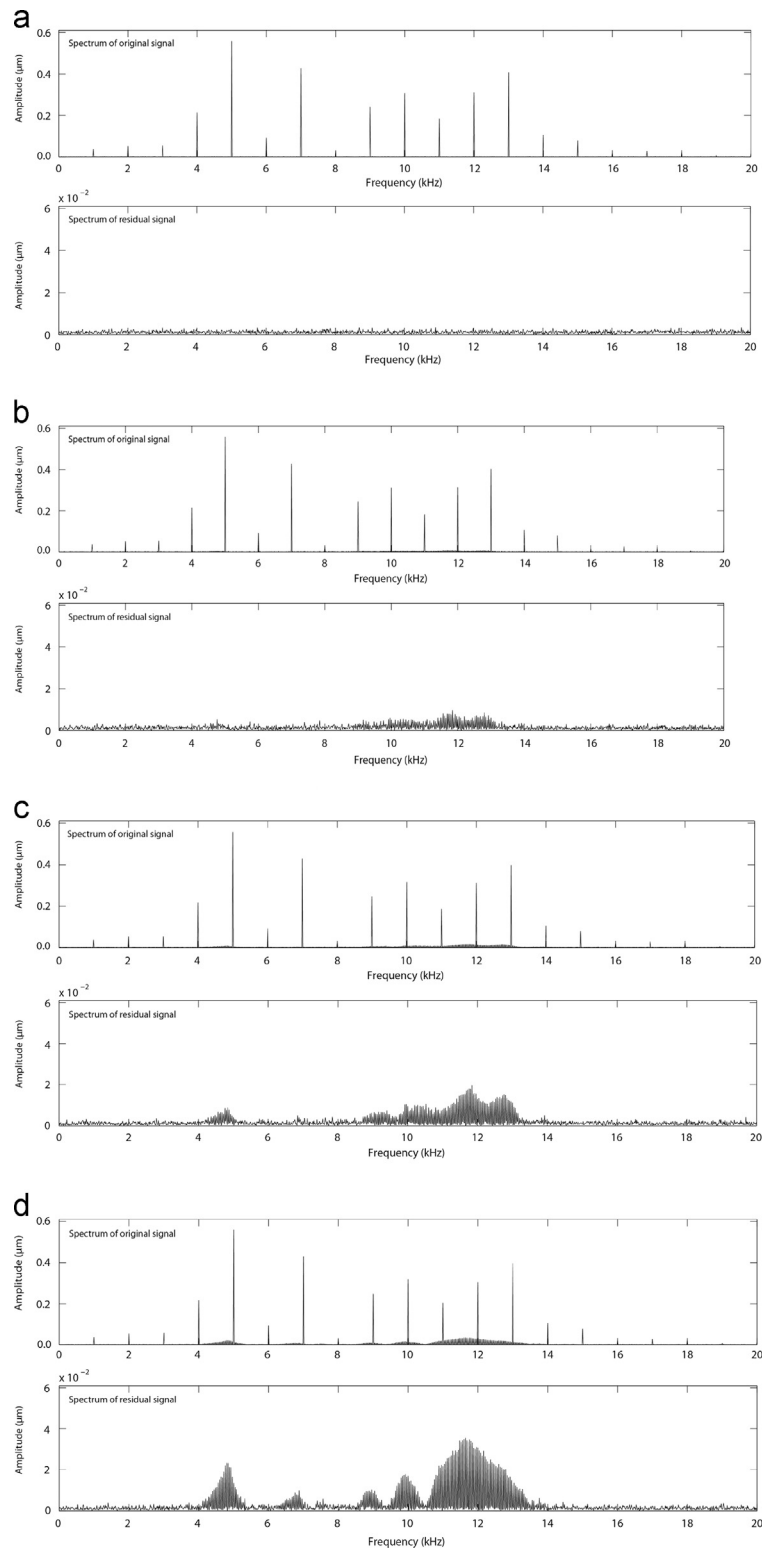


Fig. 14. The change of the spectra for selected original signals and their residual signals for the 3rd scenario. (a) Crack depth = 0.0 mm and CL = 0%, (b) crack depth = 0.8 mm and CL = 21.5%, (c) crack depth = 1.2 mm and CL = 32.25% and (d) crack depth = 1.8 mm and CL = 48.38%.

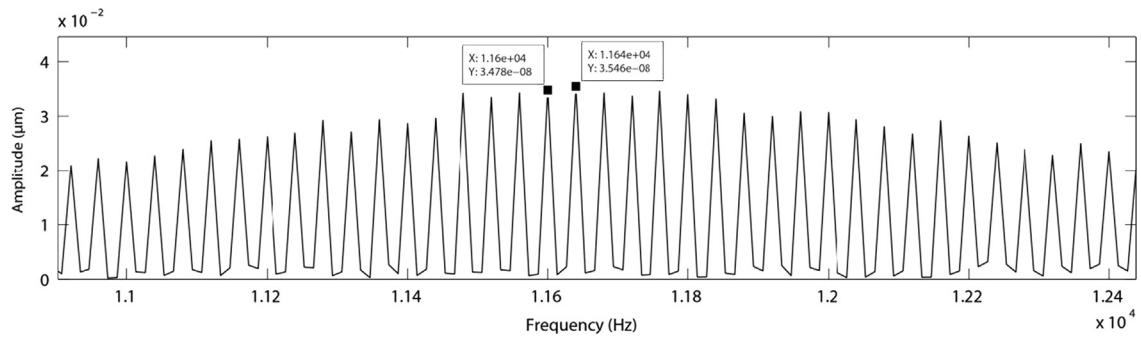


Fig. 15. Zoomed part of the spectra in Fig. 14 case (d) showing the multiple integers of the rotational speed 40 Hz.

In the case where the mean value of the signal is not zero, the RMS indicator can be obtained as follows [22]:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{n=1}^N (x(n) - \bar{x})^2} \quad (23)$$

where $\bar{x} = (1/N) \sum_{n=1}^N x(n)$.

Kurtosis is an indicator which measures the degree of peakiness of a distribution and describes the signal shape as compared to the normal distribution. The kurtosis value depends on the distribution tail length; so that the kurtosis value of the residual signal is much higher than that of the original signal, see Fig. 11. The kurtosis indicator can be defined as follows [1,22]:

$$\text{kurtosis} = \frac{1}{N} \sum_{n=1}^N \frac{(x(n) - \bar{x})^4}{\left[\frac{1}{N} \sum_{n=1}^N (x(n) - \bar{x})^2 \right]^2} \quad (24)$$

9. Results and discussion

In the mesh stiffness calculation, the assumption of using a reduced dimension when calculating I_x and A_x needs to be investigated further in order to establish which method is the most accurate when calculating the mesh stiffness analytically for gears with a cracked tooth. The stiffness will become lower with the used model and therefore the signal will show to strong influence of the crack. However, it is essential to evaluate the difference between different crack propagation scenarios against earlier studies by using the same model.

Dynamic simulation with time domain analysis has been performed to obtain the residual signals of all the studied cases for each crack propagation scenario. This work was carried out by studying the pinion's y-displacement, which is the most sensitive movement for crack propagation in the pinion tooth root.

Normally distributed noise is added with a SNR value of 20 dB to include the influence of measurements noise. In reality random manufacturing error or any other contribution which is synchronous with the rotational speed will add energies in the spectra at multiple integers of the rotational speed. This signal type will theoretically be removed from future signal content by the residual signal process. Furthermore any non-synchronous contribution can be removed or reduced by time signal averaging.

In this simulation study, a random measurement noise is applied which adds energies at all frequencies of the spectra including the integer multiples of the rotational speed. Any pseudorandom noise or signal synchronous with the rotational speed is removed in the residual signal process and therefore not considered in this simulation.

However, any constant amount of pseudorandom noise or noise with continuous spectra will only add a constant effect to the indicators RMS and kurtosis. Therefore both the measurement noise and the pseudorandom noise will have the same effect on the indicators and the comparison between the studied crack scenarios.

Fig. 11 shows the original signal of the healthy case and of the three selected faulty cases with their residual signals according to the 3rd crack propagation scenario. This figure illustrates clearly the importance of performing the fault detection analysis based on the residual signal. The statistical indicators are more sensitive to the crack size when the residual signal is utilised.

Figs. 12 and 13 show the changes in the statistical indicators for the three studied crack propagation scenarios. It is clear that the statistical indicators for the 1st scenario increase significantly, implying a dramatic decrease in the gear mesh stiffness, and this scenario can be considered as the scenario for which fault detection is most easily performed. The 2nd scenario shows the smallest increase in the statistical indicators, and this scenario can therefore be considered as the

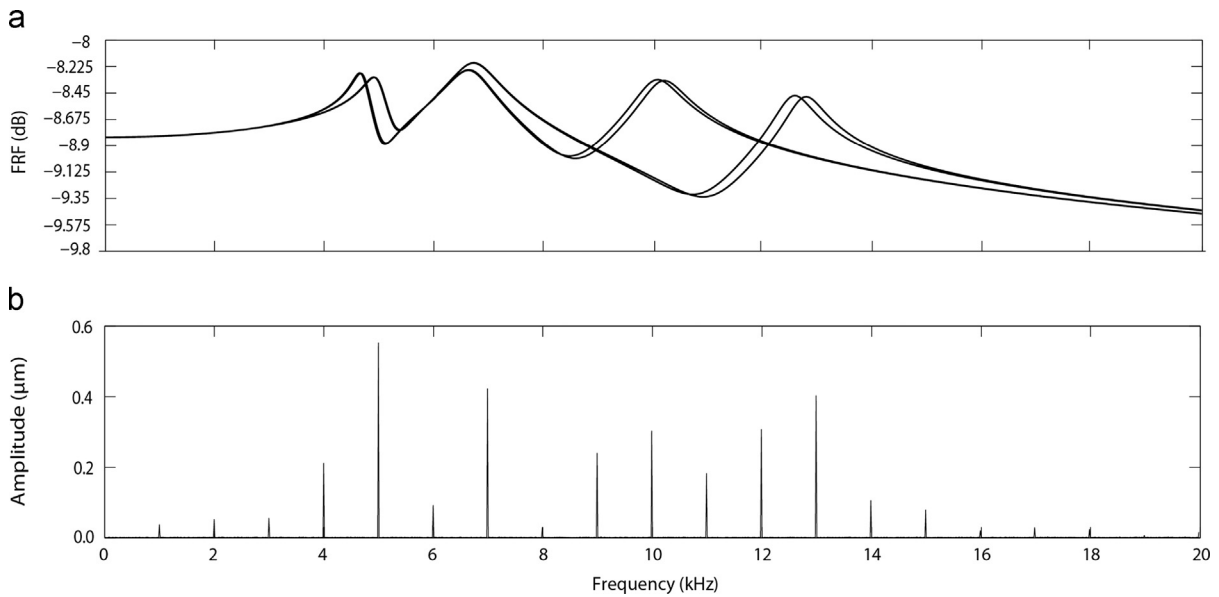


Fig. 16. (a) Frequency response functions (receptance of pinion centre) of healthy gear with four different values of mesh stiffness and (b) spectrum of original signal of healthy gear.

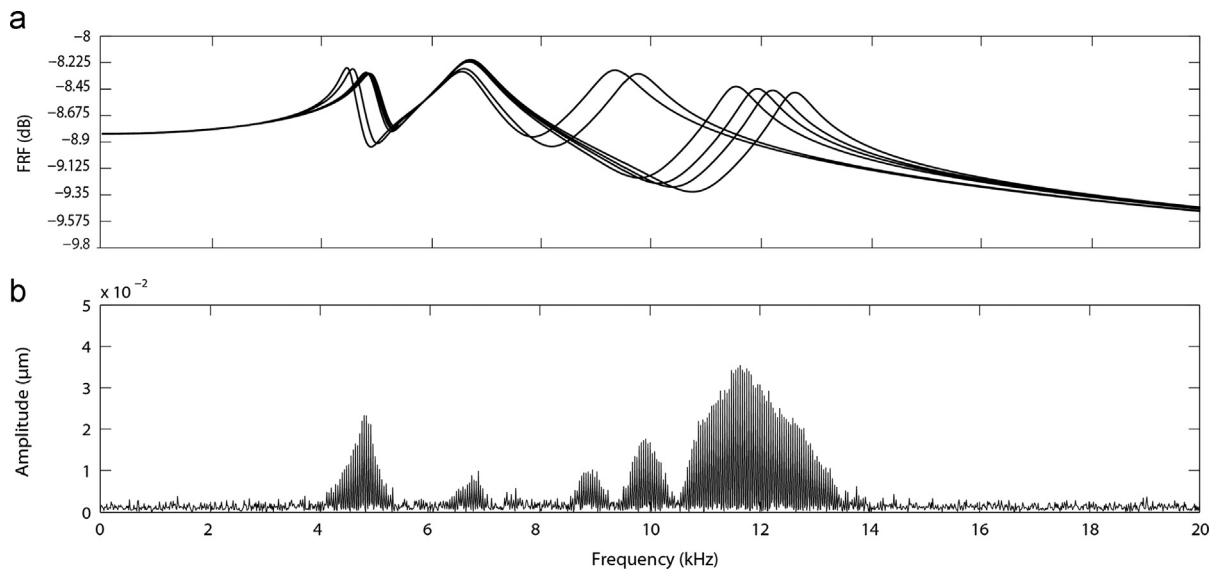


Fig. 17. (a) Frequency response functions (receptance of pinion centre) of faulty gear with six different values of mesh stiffness and (b) spectrum of residual signal of faulty signal.

hardest scenario for detecting crack propagation. The 3rd scenario shows a moderate behaviour in comparison with the other scenarios. The statistical indicators start with a slight change at the beginning of crack propagation, after which they increase at a significant rate. The fault detection for the 3rd scenario is critical as it is difficult to perform an early detection with very slight initial changes in the statistical indicators, after which these indicators show a significant change when the crack goes deeper, implying a serious crack propagation condition.

In the healthy condition the sidebands generated in the frequency spectrum are approximately constant with time. When the fault starts propagation, the number and amplitude of the sidebands start changing and increase with the crack size [40]. Therefore, the sidebands can be utilised as a good indication for fault detection. In the present research study, the spectra were obtained for all the studied cases. Fig. 14 shows a part of the spectrum of the healthy case and of the three selected faulty cases, with the spectrum of their residual signals according to the 3rd scenario. Each spectrum was created using four simulated revolutions where the first revolution including the transient response was excluded. Fig. 11 is showing

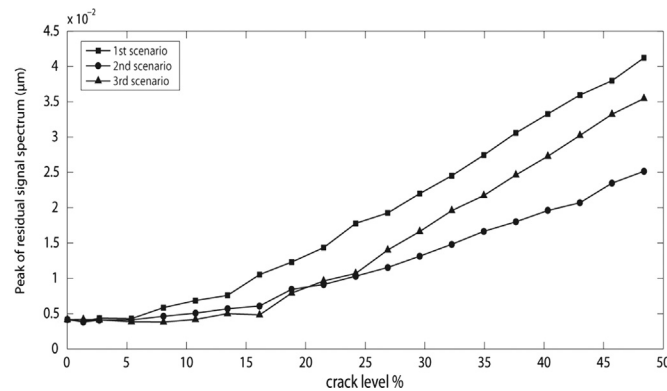


Fig. 18. Curves describing the peak increase in spectra of residual signals for the three studied scenarios.

one of the three revolutions used for calculating the spectra. The full record of 15,000 (5000 samples/revolution) was used to create the spectra without reducing the spectral resolution in the Matlab FFT command. In the figure the mesh frequency at 1 kHz and its multiples can be seen. Increasing amplitudes of the sidebands can also be seen with increasing crack depth. In Fig. 14a, a continuous spectrum can be seen due to the presence of the simulated measurement noise. In Fig. 14b–d, the contribution of the faulty gear tooth repeated for each revolution can be seen as peaks at multiple integers of the rotational speed 40 Hz. These multiple integers are illustrated in Fig. 15.

Fig. 16a shows the frequency response functions (receptance at pinion centre) of a healthy gear for four different values of gear mesh stiffness (3.55 , 3.68 , 1.99 and 2.02×10^8 N/m, see Fig. 8b). Fig. 16b shows the corresponding spectrum of the original signal. Fig. 17 illustrates the same as Fig. 16 but for a faulty gear case (48.38% crack level). For the faulty case the receptance is plotted for six different gear mesh stiffnesses (3.57 , 3.31 , 1.77 , 1.56 , 3.12 and 2.87×10^8 N/m, see Fig. 8b). The spectra shown in both the figures represent the signal produced by the dynamic system with varying gear mesh stiffness. The gear mesh frequency and its multiples are filtered through the dynamic system of each case, amplifying or attenuating the peaks according to the resonances of the dynamic system. This effect can be seen in Fig. 16 where the gear mesh peaks in graph (b) can be compared to the system resonances described by the receptance in graph (a).

Fig. 18 shows the change in the sidebands' peak amplitude for the three crack propagation scenarios. It is clear that the sidebands' amplitude for the 1st scenario increases significantly and shows the highest level. The corresponding amplitude for the 2nd scenario is lower than that for the 1st one and shows the overall lowest increase rate. The sidebands' amplitude for the 3rd scenario shows a very slight increment at the beginning of crack propagation, with a lower rate than that for the 2nd one, until a crack level around 20% when the sidebands' amplitude starts to exceed that for the 2nd scenario with a significant increment.

10. Conclusions

This study has investigated the performance of the statistical fault detection indicators (the RMS and kurtosis) in each proposed crack propagation scenario to compare these scenarios from a fault diagnostics point of view. Dynamic simulation has been performed to obtain the residual signals of all the studied cases for each crack propagation scenario. The comparison of the statistical indicators applied to the residual signals shows that in the 1st scenario, which has already been treated in many previous studies, the faults are most easily detectable, since in this scenario there is a change in the indicators implying a dramatic decrease in the gear mesh stiffness. The fault detection in the 2nd scenario is not so easy, as the crack propagates with no significant reflection on the mesh stiffness loss. The 3rd proposed scenario should receive more attention in research because it could occur in reality in case of non-uniform load distribution. However, the 3rd scenario shows a critical behaviour and it is difficult to perform early detection, since there is a very slight change in the statistical indicators at the beginning of the crack propagation, after which these indicators show a significant change when the crack goes deeper, which implies a serious crack propagation condition.

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